

Problem Solution Fit

Project: Rising Water – AI-Based Flood Prediction System

1. Problem Overview

Floods cause severe damage to lives, agriculture, infrastructure, and the environment. Existing warning systems are often delayed or rely on limited information.

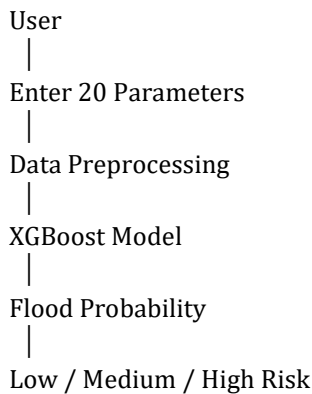
2. Proposed Solution

Rising Water is a Machine Learning-based web application that predicts flood probability using environmental parameters. Users receive a flood probability score and a risk level.

3. Problem–Solution Fit

Problem	Solution	Benefit
Late flood warnings	ML-based early prediction	More preparation time
Manual analysis	Automated prediction	Fast decision making
Complex flood factors	20 environmental inputs	Higher prediction quality
Limited accessibility	Flask web application	Easy to use
Property & crop loss	Risk categorization	Better planning

4. Workflow



5. Project Benefits

- Early flood prediction
- Supports disaster preparedness
- Easy-to-use web interface
- Reduces potential losses
- Can be extended with live weather APIs

6. Conclusion

The Rising Water project provides a practical AI-based solution that addresses the problem of delayed flood risk assessment by delivering quick and reliable predictions.